CSE STUDY HUB

Communication Skills for Engineers II — Aptitude

UNIT 5 · Logical Reasoning

What this unit is about

This unit builds understanding of **Logical Reasoning**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Direction sense: cardinal directions, turns and net displacement

2. Distance travelled vs shortest displacement

3. Shadow problems using sun direction and opposite shadow

4. Blood relations: generations, maternal/paternal links and gender clues

5. Family-tree diagrams and coded relationship statements

6. Avoiding assumptions; translate each sentence before solving

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 5 NOTES · PART 1

Direction sense: cardinal directions, turns and net displacement

Draw a compass and track each turn. Distinguish total distance travelled from net displacement. For shadow problems, the shadow points opposite the sun direction.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Distance travelled vs shortest displacement

Compare the alternatives in Distance travelled vs shortest displacement using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Logical Reasoning, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 2

Shadow problems using sun direction and opposite shadow

The OSI model has Physical, Data Link, Network, Transport, Session, Presentation and Application layers. Each layer provides services to the one above and uses the one below; encapsulation adds headers as data moves down.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Blood relations: generations, maternal/paternal links and gender clues

Convert each relationship statement into a small family tree. Mark gender only when stated, track generations carefully, and never assume a relationship that the question does not provide.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Logical Reasoning, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 3

Family-tree diagrams and coded relationship statements

A tree is a hierarchical, connected acyclic structure. Learn root, leaf, parent, child, depth and height. Traversals visit nodes in a defined order; BST ordering enables efficient search when balanced.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Avoiding assumptions; translate each sentence before solving

Avoiding assumptions; translate each sentence before solving is a syllabus concept in Communication Skills for Engineers II — Aptitude. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Logical Reasoning, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Direction sense: cardinal directions, turns and net displacement** — know the definition, core working and one example.

• **Distance travelled vs shortest displacement** — know the definition, core working and one example.

• **Shadow problems using sun direction and opposite shadow** — know the definition, core working and one example.

• **Blood relations: generations, maternal/paternal links and gender clues** — know the definition, core working and one example.

• **Family-tree diagrams and coded relationship statements** — know the definition, core working and one example.

• **Avoiding assumptions; translate each sentence before solving** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Logical Reasoning. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Direction sense: cardinal directions, turns and net displacement and explain its importance.

Q2. Explain the working/steps involved in Distance travelled vs shortest displacement.

Q3. Differentiate between two related concepts from Logical Reasoning.

Q4. Give one practical application of Avoiding assumptions; translate each sentence before solving and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Logical Reasoning.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.